

Question	Marking Guidance	Mark	Comments
09.1	1. Fetal blood has more oxygen OR Fetal blood has less carbon dioxide; 2. (Because) gas exchange occurs in the placenta OR Gas exchange does not occur in (fetal) lungs;	2 (2 x AO2)	1. Accept converse for references to mother's pulmonary artery 1. Accept fetal blood is oxygenated 1. Accept high for 'more' OR low for 'less' 1. Ignore affinity

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09.2	1. (IgG) antibodies (from mother) are complementary/bind specifically; 2. To pathogens/antigens crossing the placenta; 3. Giving passive immunity (in fetus) OR Stopping symptoms forming (in fetus) OR Giving immediate/rapid protection (in fetus);	3 (3 x AO2)	1. Accept Antibodies bind with antigens / antigen-antibody complex

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09.3	(Against measles) 1. (To achieve) herd immunity to reduce spread; (Against tetanus) 2. No herd immunity OR Skin wounds are common (in children);	2 (2 x AO2)	1. Accept 'herd effect' for herd immunity 2. Accept Only protects the individual

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09.4	Reduced vaccination (in children) OR virus has mutated;	1 (AO2)	Accept 'more unvaccinated individuals entering the country/ population' Reject disease mutated

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09.5	1. (Production of) more memory cells; 2. (So) higher concentration of antibodies (in blood) OR (So) more rapid production of antibodies (on further infection);	2 (2 x AO2)	Ignore active immunity 2. Accept More antibodies (in blood)